Arthur Week 10 - Calibration Verification

Scope

This is an independent verification layer for the ensemble calibration work. It does not run a competing grid search, change detector algorithms, or use the held-out test split. Its metric example is synthetic and hand-calculated.

Implemented safeguards

- classification_metrics treats stego as positive and reports accuracy, precision, recall, and F1. Undefined precision/recall/F1 are reported as zero.
- predictions_from_scores uses score >= threshold for stego and validates both threshold and normalized score ranges.
- validate_weights requires exactly the chi, RS, and DH weights, finite and non-negative, summing to 1.
- ensemble_score consumes normalized [0, 1] detector inputs and inverts raw RS as 1 - rs_raw before applying the weighted vote. Chi-square must be normalized to suspiciousness (1 - p_value) before this function. DH's normalized [0, 1] score is consumed directly.
- validation_rows refuses any row whose split is not explicitly validation, including held-out test rows.
- verify_selected_configuration requires the selected weights and threshold to match a row in the search results.

Hand calculation

For truth [clean, stego, stego, clean] and predictions [clean, stego, clean, clean], TP=1, TN=2, FP=0, FN=1. Expected metrics: are accuracy 3/4 = 0.75, precision 1, recall 1/2 = 0.5, and F1 2/3. A unit test checks these values directly.

Threshold boundary checks confirm 0.49 is clean, 0.50 is stego, and 0.51 is stego at threshold 0.50.

Integration status

The workspace currently contains Week 7 detector snapshots and a development score CSV. It does not contain the Week 9 ensemble implementation or Menishen's Week 10 grid-search output. Therefore the following remain pending integration checks against the official pipeline: chi-square normalization parity, actual RS direction parity with Week 9, its DH input adapter, validation metric recalculation, and membership of the official selected row. No official configuration or validation performance is claimed here.

Run the verification tests

From arthur-week7, run:

```
python -m pytest tests/test_calibration_metrics.py -q
```

The held-out dataset is not loaded by these tests.